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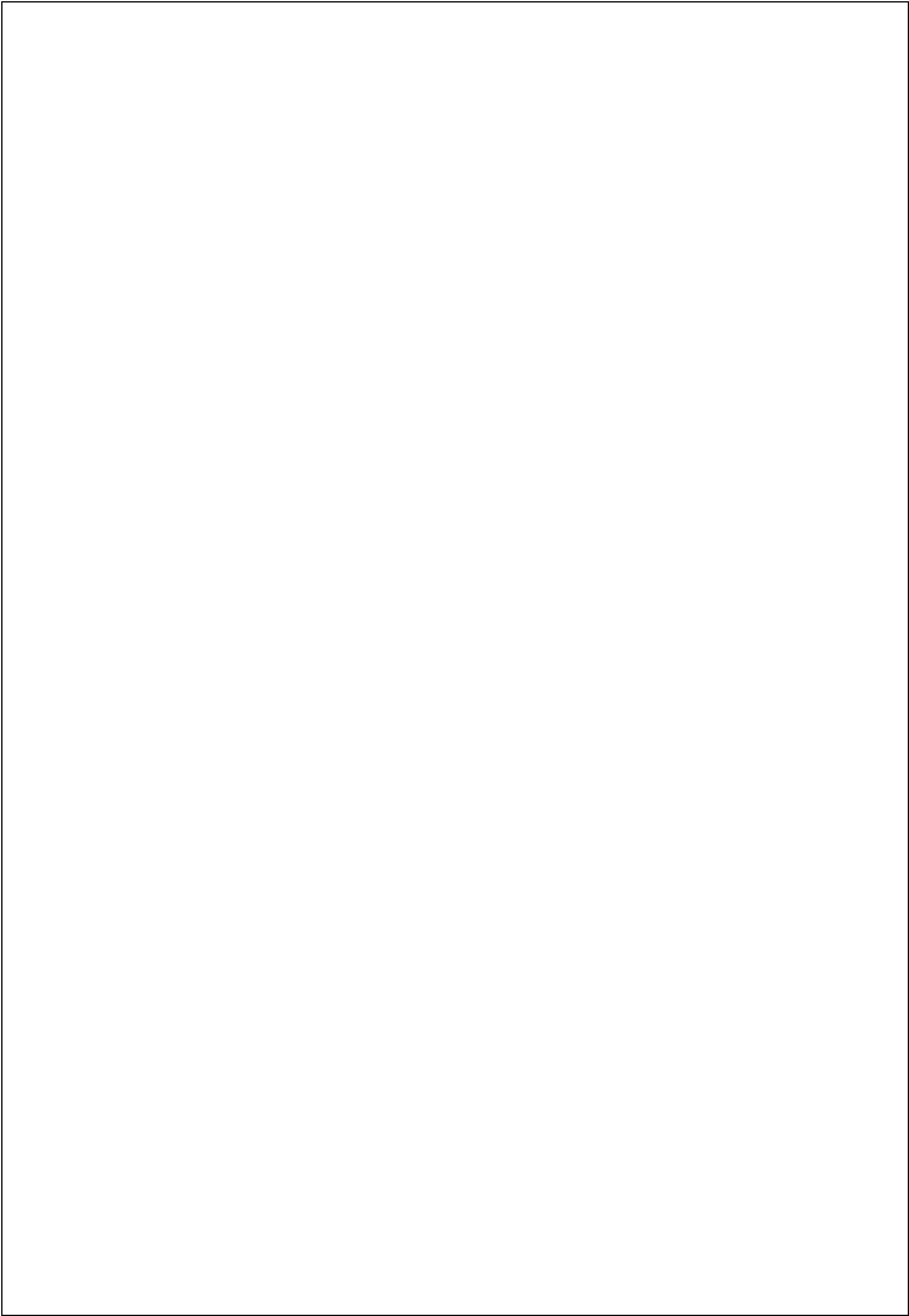
INTERNSHIP REPORT

MINI PROJECT

NAME : GOKUL R

REG. NO : 713923243005

BRANCH : B.Tech - Artificial Intelligence and Data Science





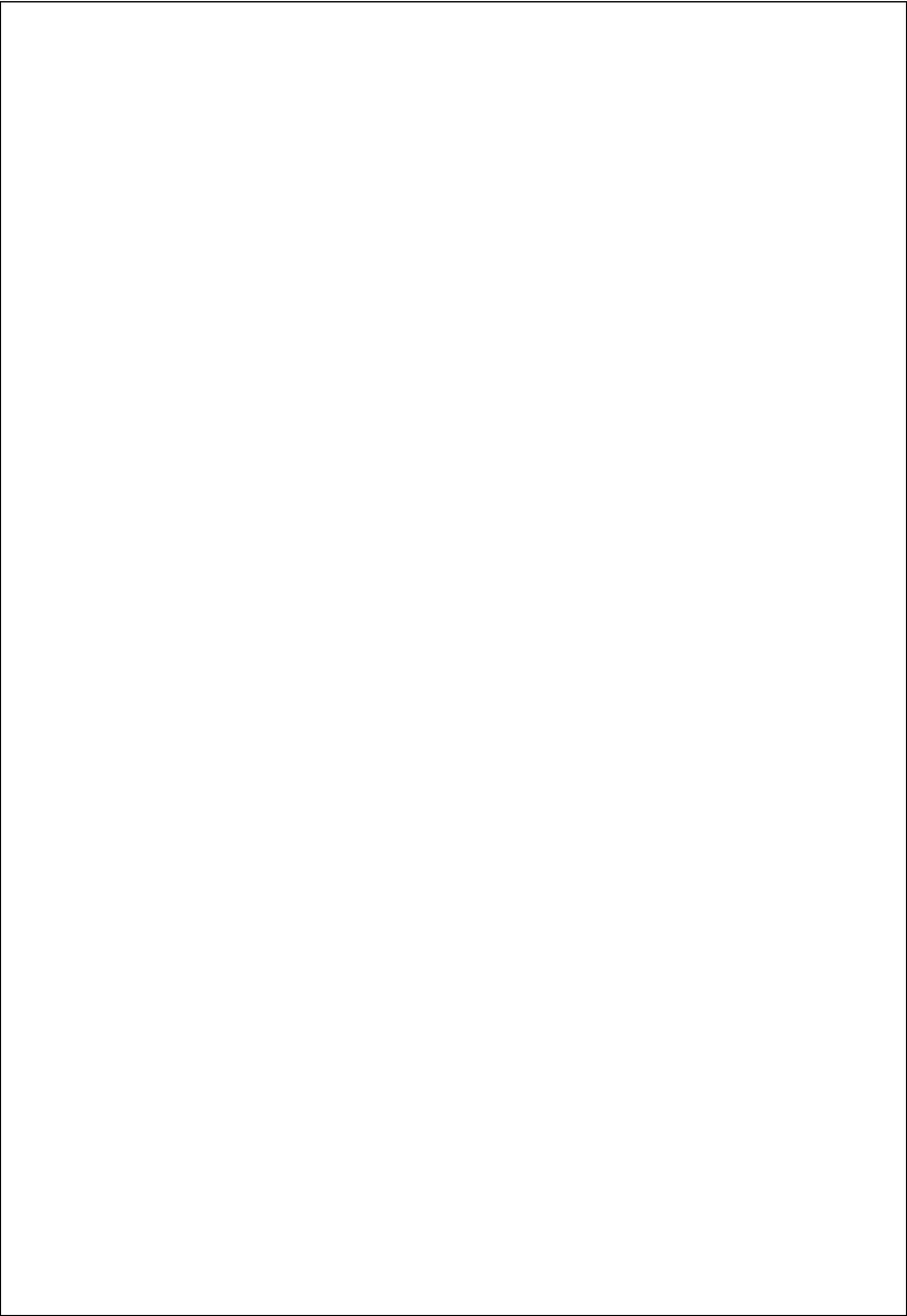
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INTERNSHIP ATTENDANCE SHEET

NAME OF INTERN		R. Gokul	ROLL.NO & BATCH	713923243005 2023 - 2027
NAME OF ORGANIZATION		Sri Ranganartha Institute of Engineering & Tech		
NAME & DESIGNATION OF INDUSTRY - SUPERVISOR		Qaroo India Private Limited Poorani. G		
S.NO	DATE	TASK AT HAND	DAILY SIGNATURE INTERN	WEEK VERIFICATION & REMARKS BY INDUSTRY - SUPERVISOR
1	15/12/25	Introduction to Python		
2	16/12/25	Conditional statements		
3	17/12/25	Loops & Compound data types		
4	18/12/25	Slicing and functions		
5	19/12/25	Compound data types		
6	20/12/25	Class & Object		
7	22/12/25	OOPS concept		
8	23/12/25	Inheritance & Polymorphism		
9	24/12/25	Encapsulation		
10	26/12/25	File Handling		
11	27/12/25	Read Files		
12	28/12/25	Write/Create files		
13	30/12/25	Delete files		
14	31/12/25	Python Libraries		
15	2/1/26	Project discussion		

NOTES: An internship attendance sheet is a vital document that ensures systematic tracking of intern participation and plays an important role in academic and professional evaluation.



Acknowledgement

I would like to take this opportunity to express my sincere gratitude to all those who have supported and guided me in the successful completion of my internship.

I am extremely thankful to **Qaroo India Private Limited, Coimbatore**, for providing me the valuable opportunity to undergo a **15-day internship in Python Programming**. This internship helped me gain practical knowledge of Python concepts and improved my understanding of real-time application development.

The training enhanced my technical skills and boosted my confidence in programming.

I express my heartfelt thanks to my **industry supervisor and mentors** at Qaroo India Private Limited for their continuous guidance, encouragement, and support throughout the internship period. Their valuable suggestions and practical explanations helped me understand Python concepts clearly and effectively.

I would also like to thank the **Management, Principal, and Faculty Members of Sri Ranganathar Institute of Engineering and Technology** for their constant support and for providing this internship as a part of the academic curriculum.

Finally, I thank my family and friends for their motivation and encouragement, which helped me complete this internship successfully .

Gokul R

Sri Ranganathar Institute of Engineering and Technology

Athipalayam

Declaration for the Internship Report

I, **R. Gokul**, a student of **Sri Ranganathar Institute of Engineering and Technology**, hereby declare that this **Internship Report** titled **“Python Programming Internship”** is a bonafide record of the work carried out by me during my **15-day internship at Qaroo India Private Limited, Coimbatore**.

I further declare that this report has been prepared by me under the guidance of the concerned supervisors and mentors of Qaroo India Private Limited. The work presented in this report is original and has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate.

All the information, data, and observations mentioned in this report are true to the best of my knowledge and belief. I take full responsibility for the correctness and authenticity of the contents of this internship report.

Place: _____

Date: _____

Signature: _____

(Gokul R)

Sri Ranganathar Institute of Engineering and Technology

Athipalayam

Certificate from the supervisor

Table of Content

S.NO	TOPIC
1	Duration /no. Of days
2	Introduction
3	Prolife of organization
4	Activities undertaken
5	Tools and techniques of social work applied
6	Observation
7	Evaluation
8	Learning
9	Output
10	Reflection
11	Conclusion
12	Certificate

Duration /no. Of days:

Day 1 – Introduction to Python

On the first day, I was introduced to the Python programming language and its history. The trainer explained the features of Python such as simplicity, readability, and platform independence. I learned about different applications of Python in areas like web development, data analysis, and automation. Basic syntax and execution of simple Python programs were demonstrated.

Day 2 – Conditional Statements

On the second day, I learned about decision-making statements in Python. The concepts of `if`, `elif`, and `else` were explained in detail. I practiced writing programs that execute different blocks of code based on conditions. This session helped me understand logical thinking in programming.

Day 3 – Looping Conditions

On the third day, I studied looping concepts such as `for` and `while` loops. The trainer explained how loops are used to perform repetitive tasks efficiently. Loop control statements like `break` and `continue` were also discussed. Practical examples were solved during the session.

Day 4 – Strings and Functions

On the fourth day, I learned about strings and their operations such as slicing, concatenation, and formatting. I also studied functions and their importance in modular programming. I practiced writing user-defined functions and passing parameters to them. This helped in organizing code effectively.

Day 5 – Compound Data Types

On the fifth day, I learned about Python data structures such as lists, tuples, sets, and dictionaries. The trainer explained their properties and differences. I practiced performing operations like insertion, deletion, and searching. This session helped me manage multiple values efficiently.

Day 6 – Object Oriented Programming (OOP) Concepts

On the sixth day, I was introduced to Object Oriented Programming in Python. I learned about classes, objects, constructors, and methods. The importance of OOP in real-world applications was explained. Simple programs were written to understand class-based programming.

Day 7 – Practical Session

On the seventh day, a full practical session was conducted. I practiced Python programs covering conditions, loops, functions, and data structures. The trainer clarified doubts and guided me in correcting errors. This session improved my confidence in coding.

Day 8 – Inheritance and Polymorphism

On the eighth day, I learned about inheritance and polymorphism in Python. The concept of code reusability using inheritance was explained. I practiced writing programs with parent and child classes. Polymorphism was demonstrated using method overriding.

Day 9 – Encapsulation

On the ninth day, I studied encapsulation and data hiding in Python. I learned how access modifiers are used to protect data within a class. This concept helped me understand secure and controlled access to class members. Practical examples were discussed.

Day 10 – File Handling

On the tenth day, I learned the basics of file handling in Python. The trainer explained different file modes such as read, write, and append. I practiced opening and closing files properly. The importance of file handling in real-time applications was discussed.

Day 11 – Reading Files

On the eleventh day, I learned how to read data from files using Python. Different file reading methods were explained. I practiced reading text files line by line. This session helped me understand file data processing.

Day 12 – Writing and Creating Files

On the twelfth day, I learned how to create new files and write data into them using Python. I practiced writing programs to store user input into files. File handling errors and exceptions were also discussed briefly.

Day 13 – Deleting Files

On the thirteenth day, I learned how to delete files using Python. The importance of proper file management was explained. I practiced deleting files safely using Python programs. This session helped me understand system-level operations.

Day 14 – Python Libraries

On the fourteenth day, I was introduced to Python libraries and modules. The trainer explained the importance of libraries in reducing development time. I learned about commonly used libraries and how to import and use them in programs. Simple examples were demonstrated.

Day 15 – Project Discussion

On the final day, a project discussion was conducted. The trainer explained the project objectives and workflow. We discussed how Python concepts learned during the internship can be applied to real-world projects. This session helped me understand practical implementation and project planning.

INTRODUCTION

In the present digital era, programming skills play a vital role in the development of software applications, automation systems, and data-driven technologies. Among various programming languages, **Python** has emerged as one of the most popular and widely used languages due to its simplicity, versatility, and powerful capabilities. Python is an interpreted, high-level programming language known for its clear syntax, readability, and extensive library support, making it suitable for beginners as well as professionals.

Python is extensively used in diverse fields such as **web development, data science, artificial intelligence, machine learning, automation, cybersecurity, and software testing**. Its ability to integrate with other technologies and platforms has made it a preferred choice in both academic and industrial environments. Learning Python not only strengthens logical and analytical thinking but also provides a strong foundation for understanding advanced technologies and real-world software development practices.

To enhance my practical knowledge and gain hands-on experience in Python programming, I successfully completed a **15-day internship in Python Programming at Qaroo India Private Limited, Coimbatore**. This internship was designed to provide a balanced combination of theoretical concepts and practical implementation. The training began with the fundamentals of Python and gradually progressed to advanced topics such as object-oriented programming, file handling, and the use of Python libraries.

During the internship, I learned essential programming concepts including conditional statements, looping constructs, functions, string manipulation, and compound data types. I was also introduced to object-oriented principles such as classes, objects, inheritance, polymorphism, and encapsulation, which are crucial for developing structured and reusable code. In addition, file handling techniques such as reading, writing, creating, and deleting files were covered to understand data storage and management in Python applications.

The internship also emphasized practical learning through hands-on sessions, exercises, and project discussions. These activities helped me apply the concepts learned in real-time scenarios and improved my problem-solving and debugging skills.

Exposure to Python libraries further enhanced my understanding of how Python simplifies development and improves efficiency in application design.

This internship report presents a detailed account of the organization, internship duration, daily activities, tools and technologies used, observations, learning outcomes, and overall experience gained during the training period. The internship significantly improved my technical knowledge, programming skills, and confidence in Python, thereby preparing me for future academic projects and professional opportunities in the field of software development and emerging technologies.

PROFILE OF THE ORGANIZATION

Qaroo Learnings India Pvt Ltd, also known as Qaroo Technologies India Pvt Ltd, is a reputed training and software development organization located in Coimbatore, Tamil Nadu. The company is well known for delivering quality technical education and skill-oriented training programs tailored to meet current industry standards. It primarily focuses on emerging and in-demand technology domains such as Data Science, Artificial Intelligence, Machine Learning, Database Management Systems, and Software Development.

The organization's core objective is to bridge the gap between academic learning and industry requirements by providing a strong foundation in both theoretical concepts and practical implementation. Qaroo Learnings emphasizes hands-on training, real-time project exposure, and structured learning modules, which enable students to gain industry-ready skills. The training programs are designed to enhance problem-solving abilities, logical thinking, and technical proficiency among learners.

Qaroo Learnings offers both online and offline training modes, supported by modern infrastructure, well-equipped computer laboratories, and advanced learning resources. The company follows a learner-centric approach and regularly updates its curriculum based on evolving technological trends and market demands. It is recognized by UGC and NSDC, which adds significant value and credibility to its training and certification programs.

One of the key strengths of Qaroo Learnings is its 100% placement assistance, which includes resume preparation, mock interviews, aptitude training, and career counseling. In addition to technical skills, the organization places strong emphasis on communication skills, professional ethics, and personality development. The faculty members consist of experienced industry professionals and certified trainers who provide continuous mentorship and guidance throughout the training period.

Overall, Qaroo Learnings India Pvt Ltd has earned positive recognition for its supportive learning environment, practical teaching methodology, and student-focused approach, making it a trusted platform for aspiring professionals seeking to build successful careers in the technology sector.

ACTIVITIES UNDERTAKEN DURING THE INTERNSHIP

During the 15-day Python internship at **Qaroo India Private Limited, Coimbatore**, I actively participated in various learning activities designed to strengthen both theoretical knowledge and practical programming skills. The internship followed a systematic and hands-on approach, enabling gradual learning from basic concepts to advanced Python programming techniques.

Initially, the internship focused on understanding the fundamentals of Python programming. I learned about Python syntax, keywords, data types, and basic input-output operations. Simple programs were written and executed to gain familiarity with the Python environment.

I also learned **string operations and user-defined functions**, which helped in understanding modular programming and code reusability. Activities included writing functions with parameters, return values, and performing string manipulations such as slicing, formatting, and searching. This enhanced code organization and readability.

The internship further covered **compound data types** including lists, tuples, sets, and dictionaries. I practiced storing, accessing, modifying, and managing collections of data efficiently.

A major part of the internship was dedicated to **Object Oriented Programming (OOP)** concepts. I learned about classes, objects, constructors, inheritance, polymorphism, and encapsulation. Practical programs were developed to implement these concepts, which helped in understanding structured programming and real-world software design principles.

The internship also included extensive training in **file handling operations**. I learned how to create, read, write, append, and delete files using Python. Practical exercises were performed to manage data stored in files, which helped me understand data persistence and system-level operations.

Towards the end of the internship, I was introduced to **Python libraries** and their importance in simplifying development tasks. I learned how to import and use libraries to enhance functionality and improve efficiency. A project discussion was conducted where the application of Python concepts in real-world scenarios was explained.

TOOLS AND TECHNIQUES OF SOCIAL WORK APPLIED

Although the internship was primarily focused on technical training in Python programming, several tools and techniques of social work and professional practice were applied throughout the internship period.

Time management was an important technique practiced during the internship. Daily tasks, programming exercises, and practice sessions were completed within the given time frame. Proper planning helped in balancing learning activities and practical implementation efficiently.

Problem-solving and analytical thinking were applied while writing, testing, and debugging Python programs. Logical reasoning was used to identify errors, understand program flow, and improve code efficiency.

Communication and interpersonal skills were strengthened through regular interaction with mentors and peers. Doubts related to Python concepts were discussed clearly, and feedback was received constructively.

Teamwork and collaborative learning played a key role during practical sessions and project discussions. Sharing ideas, helping peers, and learning from others promoted a cooperative learning environment and enhanced mutual understanding.

Ethical responsibility and discipline were followed while writing programs and handling data. Proper coding practices were maintained, and originality in work was ensured. Respect for rules, punctuality, and professional behavior were observed throughout the internship.

Self-learning and adaptability were also practiced by exploring additional Python concepts, experimenting with programs, and referring to documentation when needed.

From a technical perspective, **Python programming tools**, file handling techniques, and Python libraries acted as effective learning tools.

Overall, the application of these social work tools and techniques enhanced both technical competence and personal development, preparing me for professional and ethical responsibilities in future academic and industry environments.

OBSERVATIONS

During the 15-day Python internship at **Qaroo India Private Limited, Coimbatore**, several important observations were made regarding programming practices, learning approaches, and real-time application development.

One of the key observations was that **Python is a highly user-friendly and efficient programming language**, especially for beginners. Its simple syntax and readability make it easy to understand programming logic and implement solutions quickly. This helps reduce development time and minimizes coding errors.

Another important observation was the effectiveness of **Object Oriented Programming (OOP)** concepts in building structured and reusable code. Using classes, objects, inheritance, and encapsulation helps organize programs better and makes them easier to maintain and extend in real-world applications.

The internship highlighted the importance of **hands-on practice**. Practical sessions played a major role in strengthening theoretical knowledge. Writing programs, debugging errors, and experimenting with code helped in understanding concepts more deeply than theory alone.

It was also observed that **file handling operations** are crucial in real-world applications where data needs to be stored, retrieved, and managed efficiently. Reading, writing, creating, and deleting files using Python demonstrated how data persistence is handled in software systems.

The use of **Python libraries** showed how development can be simplified and made more efficient. Libraries reduce the need for writing lengthy code and help implement complex functionality easily, which is important in modern software development.

Overall, the internship provided a clear understanding of how **theoretical concepts are applied practically**, emphasizing discipline, accuracy, and continuous learning. The observations gained during the internship contributed significantly to improving my technical knowledge, confidence, and readiness for real-world programming challenges.

EVALUATION

The evaluation of the internship was carried out through continuous observation, practical assessments, and overall performance during the training period. The internship was designed to assess both theoretical understanding and practical application of Python programming concepts.

Throughout the internship, regular practice sessions and programming exercises were conducted to evaluate my understanding of basic and advanced Python concepts. Topics such as conditional statements, looping constructs, functions, data structures, and object-oriented programming were assessed through hands-on coding tasks. This approach ensured gradual improvement and reinforced learning effectively.

A significant part of the evaluation was based on **practical performance**. Writing Python programs, debugging errors, and implementing logical solutions played a major role in measuring progress. The ability to analyze problems, apply appropriate logic, and produce correct output was continuously monitored by the mentors.

Participation in **practical sessions and project discussions** was also considered during the evaluation process. Active involvement, timely completion of tasks, and responsiveness to feedback demonstrated commitment and willingness to learn. Feedback provided by the trainers helped in identifying strengths and areas that required improvement.

File handling exercises and the use of Python libraries were evaluated to understand real-world application skills. The ability to read, write, create, and manage files efficiently showed a clear understanding of data handling concepts in Python.

Overall, the internship evaluation confirmed that the objectives of the training were successfully achieved. The structured learning approach, continuous assessment, and hands-on practice significantly improved my Python programming skills and prepared me for future academic projects and professional responsibilities.

LEARNING

Python Programming Basics: I learned the fundamentals of Python programming, including syntax, data types, and basic input-output operations. This helped me understand how Python programs are structured and executed.

Logic Building: Through conditional statements and looping constructs, I improved my logical thinking and problem-solving skills. These concepts helped in controlling the flow of programs effectively.

Functions and Data Structures: I learned how to use functions for modular programming and worked with data structures such as lists, tuples, sets, and dictionaries to manage data efficiently.

Object Oriented Programming: I gained knowledge of Object Oriented Programming concepts like classes, objects, inheritance, polymorphism, and encapsulation. These concepts helped me understand structured and reusable code design.

File Handling: I learned file handling operations such as creating, reading, writing, and deleting files using Python. This helped in understanding data storage and management.

Python Libraries: I was introduced to Python libraries and learned how they simplify programming tasks and improve development efficiency.

Professional Skills: The internship helped me develop time management, communication, teamwork, and self-learning skills, which are important for professional growth.

PROJECT DESCRIPTION – PHISHING AND URGENCY

EMAIL DETECTOR

The **Phishing and Urgency Email Detector** is a cybersecurity-oriented project designed to automatically identify fraudulent and malicious emails that attempt to deceive users into revealing sensitive information. Phishing emails are one of the most common cyberattack methods, where attackers impersonate trusted organizations and use fear or urgency-based language to manipulate users into clicking harmful links, downloading malware, or sharing personal and financial details. This project focuses on detecting such threats by analyzing the content and patterns within emails.

The system uses **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques to examine email text, including the subject and body. It identifies suspicious keywords, unusual language patterns, and urgency phrases such as “Act now,” “Verify immediately,” or “Account will be suspended.” The detector processes emails through steps like text cleaning, tokenization, stop-word removal, and feature extraction using methods like TF-IDF. These features are then used to train a classification model that categorizes emails as *phishing*, *urgency-based scam*, or *legitimate*.

The main aim of the project is to enhance email security and protect users from cyber fraud. By providing early warnings and automatic detection, the system helps reduce the risk of data breaches and financial loss. This project also demonstrates the practical application of AI in cybersecurity and highlights how intelligent systems can improve digital safety. Overall, the Phishing and Urgency Email Detector serves as a preventive tool that promotes secure communication and increases user awareness about online threats.

1. Importing Required Libraries

Purpose: To load all required tools for text processing and machine learning.

Program:

```
import pandas as pd
```

```
import re
```

```
import nltk
```

```
from nltk.corpus import stopwords

from nltk.stem import PorterStemmer


from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
```

2. Download NLTK Data

Purpose: Stopwords help remove common words like *the*, *is*, *and* that don't add meaning.

Program:

```
nltk.download('stopwords')
```

3. Load Dataset

Purpose: We need email data labeled as phishing or safe.

Program:

```
data = pd.read_csv("emails.csv")

print(data.head())
```

4. Text Cleaning

Purpose: To clean emails by removing special characters, converting to lowercase, removing stopwords, stemming words

Program:

```
stemmer = PorterStemmer()

stop_words = set(stopwords.words('english'))

def clean_text(text):

    text = re.sub('[^a-zA-Z]', ' ', text)
```

```
text = text.lower()

words = text.split()

words = [stemmer.stem(w) for w in words if w not in stop_words]

return " ".join(words)


data['clean_text'] = data['text'].apply(clean_text)
```

5. Urgency Detection Module

Purpose: To detect urgency phrases commonly used in scams

Program:

```
urgency_words = [

    "urgent", "immediately", "act now",

    "limited time", "verify now",

    "suspend", "last chance", "alert"

]


def urgency_flag(text):

    for word in urgency_words:

        if word in text.lower():

            return 1

    return 0


data['urgency'] = data['text'].apply(urgency_flag)
```

6. Convert Text to Numerical Features

Purpose: ML models understand numbers, not text.

TF-IDF converts words to numeric importance scores.

Program:

```
vectorizer = TfidfVectorizer(max_features=3000)
X_text = vectorizer.fit_transform(data['clean_text']).toarray()
```

7. Combine Urgency Feature

Purpose: Add urgency as an extra signal to improve detection.

Program:

```
import numpy as np

X = np.hstack((X_text, data[['urgency']].values))

y = data['label']
```

8. Split Data

Purpose: Separate data into training and testing sets.

Program:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

9. Train Model

Purpose: Train a classifier to recognize phishing patterns.

Program:

```
model = MultinomialNB()
model.fit(X_train, y_train)
```

10. Evaluate Model

Purpose: Check how accurate the detector is.

Program:

```
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

11. Test on New Email

Purpose: To predict if a new email is phishing.

Program:

```
def predict_email(email):
    cleaned = clean_text(email)
    vec = vectorizer.transform([cleaned]).toarray()

    urg = urgency_flag(email)
    final = np.hstack((vec, [[urg]]))

    pred = model.predict(final)

    if pred[0] == 1:
        print("Phishing Email Detected!")
    else:
        print("Legitimate Email")

# Example
predict_email("Urgent! Your account will be suspended. Verify now!")
```

CODE EXPLANATION

1. Importing Required Libraries

- pandas → used to load and handle dataset tables
- re → used for text cleaning (removing symbols, numbers)
- nltk → NLP library for text processing
- stopwords → removes common words (the, is, and)
- PorterStemmer → reduces words to root form (e.g., verifying → verify)

2. Downloading Stopwords

- Downloads English stopwords list from NLTK
- Needed to remove meaningless words from emails

3. Loading Dataset

- Loads the dataset file
- Dataset must contain:
 - text → email content
 - label → 0 (safe), 1 (phishing)

4. Text Cleaning Function

- Removes numbers & symbols
- Converts text to lowercase
- Splits text into words
- Removes stopwords
- Applies stemming
- Joins words back into sentence

5. Urgency Detection Module

- List of scam keywords used to create pressure on users
- Phishing emails often use urgency tactics

6. TF-IDF Vectorization

- Converts text into numeric values
- TF-IDF measures importance of words

7. Combining Urgency Feature

- Adds urgency flag to text features
- Makes detection smarter

8. Splitting Data

- 80% data → training
- 20% data → testing

9. Training Model

- Uses Naive Bayes classifier
- Learns phishing patterns from data

10. Model Evaluation

- Checks how correct predictions are
- Example:
- 0.95 = 95% accuracy

11. Prediction Function

- Cleans text
- Converts to TF-IDF numbers
- Checks urgency
- Combines features
- Predicts phishing or safe

Final Output:

- Phishing detected
- Legitimate email

OUTPUT

colab Untitled1.ipynb

```
Enter email subject (or type 'exit'): IT IS NEEDED
Email Subject: IT IS NEEDED
Phishing Detection: PHISHING
Urgency Level: Normal
Enter email subject (or type 'exit'): IT SHOULD BE VERIFIED NOW
Email Subject: IT SHOULD BE VERIFIED NOW
Phishing Detection: PHISHING
Urgency Level: High
Enter email subject (or type 'exit'): VERIFY YOUR ACCOUNT
Email Subject: VERIFY YOUR ACCOUNT
Phishing Detection: PHISHING
Urgency Level: High
Enter email subject (or type 'exit'): URGENT ACTION REQUIRED
Email Subject: URGENT ACTION REQUIRED
Phishing Detection: PHISHING
Urgency Level: High
Enter email subject (or type 'exit'): PROJECT DISCUSSION REMINDER
Email Subject: PROJECT DISCUSSION REMINDER
Phishing Detection: LEGITIMATE
Urgency Level: Normal
Enter email subject (or type 'exit'): CLAIM YOUR OFFER NOW
Email Subject: CLAIM YOUR OFFER NOW
Phishing Detection: PHISHING
Urgency Level: High
Enter email subject (or type 'exit'): SCHEDULE FOR TOMORROW
Email Subject: SCHEDULE FOR TOMORROW
```

```
Email Subject: CLAIM YOUR OFFER NOW
Phishing Detection: PHISHING
Urgency Level: High
Enter email subject (or type 'exit'): SCHEDULE FOR TOMORROW
Email Subject: SCHEDULE FOR TOMORROW
Phishing Detection: LEGITIMATE
Urgency Level: Normal
Enter email subject (or type 'exit'): EXIT
```

The output of the **Phishing and Urgency Email Detector** is a clear classification result that informs the user whether an email is legitimate, a phishing attempt, or an urgency-based scam. After analyzing the email's subject and content using natural language processing and machine learning techniques, the system displays a simple message such as “**Legitimate Email**” if no suspicious patterns or urgency phrases are found. If the email contains phishing-related keywords, suspicious language, or deceptive patterns commonly used by attackers, the system alerts the user with a warning like “Phishing Email Detected.” Additionally, if the email includes pressure tactics such as “act now,” “verify immediately,” or “account will be suspended,” the detector flags it as an urgency-based scam. During testing and evaluation, the system may also show an accuracy score that represents how well the model performs in classifying emails. Overall, the output helps users quickly understand the risk level of an email, enabling them to avoid potential cyber threats, fraud, and data breaches while promoting safer digital communication.

REFLECTION

The 15-day Python internship was a valuable and enriching learning experience that significantly enhanced my technical and professional skills. Through this internship, I gained practical exposure to Python programming and understood how theoretical concepts are applied in real-world applications.

Working on the **Phishing and Urgency Email Detector** project was a valuable learning experience that enhanced both my technical and problem-solving skills. Through this project, I gained practical exposure to applying Python, machine learning, and natural language processing techniques to a real-world cybersecurity problem. I learned how phishing attacks exploit human psychology, especially through urgency and fear-based language, and how technology can be used to detect such threats.

During the development process, I improved my understanding of data preprocessing, text cleaning, feature extraction using TF-IDF, and model training using classification algorithms. Implementing the urgency detection module helped me realize the importance of combining domain knowledge with machine learning for better results. I also learned how to evaluate model performance and interpret accuracy metrics.

This project strengthened my ability to think analytically, work with datasets, and build practical AI solutions. It also increased my awareness of cybersecurity issues and the importance of protecting users from online scams. Overall, the project was a meaningful part of my internship as it allowed me to apply theoretical knowledge to a real-life application, boosting my confidence and interest in AI and cybersecurity domains.

CONCLUSION

The 15-day Python internship provided a strong foundation in Python programming and practical application development. Through structured training sessions and hands-on practice, I gained a clear understanding of core Python concepts such as conditional statements, loops, functions, data structures, object-oriented programming, and file handling.

The **Phishing and Urgency Email Detector** project was successfully developed to address the growing problem of email-based cyber threats. By using machine learning and natural language processing techniques, the system can effectively analyze email content and identify phishing attempts and urgency-based scams. The project demonstrated how intelligent systems can assist in detecting suspicious patterns, keywords, and manipulation tactics commonly used by attackers.

Through this project, the practical application of concepts such as text preprocessing, feature extraction, and classification algorithms was clearly understood. The model's ability to classify emails as phishing or legitimate shows the potential of AI in improving cybersecurity and protecting users from fraud and data breaches.

Overall, this project highlights the importance of integrating AI-driven solutions in digital communication systems to enhance security and user awareness. It serves as a strong foundation for future improvements such as real-time email integration, advanced deep learning models, and multilingual support. The successful completion of this project marks a valuable learning experience and contributes to developing safer and smarter email security solutions.

CERTIFICATE



CERTIFICATE OF COMPLETION



INTERNSHIP

Python Internship

AWARDED TO

Mr.GOKUL R

AT

Qaroo Learnings India Private Limited

DURING

15/12/2025 to 02/01/2026

GRADE A+

A green ink signature of the Managing Director.

MANAGING DIRECTOR



10/01/2026
Date of Issue



IT - ITes SSC
NASSCOM



No.3A, 3rd Floor, Sri Ram Towers, Near Lakshmi Mills Signal,
Opposite to Lulu Hypermarket, Papanayakanpalayam, Coimbatore-641037